

Mathematics 2025/26, Homework 1
bonus: 2 points, deadline: March 9, 2026

1. Consider the matrices

$$A = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 1 & 5 \\ 1 & 0 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 1 \\ 2 & 3 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}.$$

- (a) Calculate C^2
- (b) Calculate $B(C - 2I)^T$, where I is the identity matrix.
- (c) Consider the following products:

$$AB, BA, BC, CB, B^T C, CB^T, AB^T, B^T A.$$

Decide which of them can be calculated and explain why the other ones can not be calculated.

2. Write the definition of the rank of a matrix. Give an example of a 3×3 matrix such that

- (a) the rank is 1,
- (b) the rank is 2.

3. Decide whether the following vectors are linearly dependent or independent. Give reasons for your answer.

(b) $\vec{a} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}, \vec{b} = \begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix}.$

(b) $\vec{a} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \vec{b} = \begin{pmatrix} 2 \\ 6 \\ 4 \end{pmatrix}, \vec{c} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}.$

(c) $\vec{a} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \vec{b} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \vec{c} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}.$

(d) $\vec{a} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \vec{b} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \vec{c} = \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix}.$

Instructions for writing homework:

- Write your homework with solution (not only the results).
- Take a photo of the homework and convert the picture to PDF (use <https://tools.pdf24.org/en/jpg-to-pdf>). Alternatively, use a different way to make a PDF file with your homework.
- Compress the file if it is large (use <https://tools.pdf24.org/en/compress-pdf>).
- Upload the final PDF file in “cousework submissions” in UIS.